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Intelligent ventilation-on-demand control system for the construction of underground tunnel complex

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ABSTRACT

Traditional ventilation methods consume excessive energy but still fail to meet requirements in underground tunnel group construction. Thus, a closed-loop intelligent control system for ventilation-on-demand (VOD) was developed. To address dynamic changes in ventilation load and reduce energy consumption, firstly, the developed system calculates the real-time ventilation load and establishes a ventilation-network-based control mode to represent the ventilation system structure. The deep deterministic policy gradient (DDPG) method was then employed for the closed-loop control ensuring the required air volume in each branch of tunnel groups while minimizing energy consumption. After that, the developed closed-loop intelligent ventilation control system encompasses comprehensive perception, real analysis, real-time control, and continuous optimization. This system treats decision-making, control, and feedback as subsystems that reflect the adaptability between ventilation efficiency, construction progress, and power consumption. Finally, the end-edge-cloud-based software of the system was developed to enable remote control and display on large screens, personal computers (PCs), and mobile applications (Apps) to ensure precise and timely operation. The system was employed in tunnel group under construction at the Xulong Hydropower Station in Southwestern China, and the obtained results validate its advanced closed-loop control based on reinforcement learning (RL) and confirm its feasibility in engineering practice.

1 Introduction

The construction of Large underground tunnel group are one of the most dangerous work environments because of the excessive dust and hazardous gasses, inaccurate and untimely ventilation monitoring systems, and unexpected failures of ventilation systems. Ventilation systems provide fresh air, remove harmful gasses and dust, and

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maintain proper temperature and humidity. However, they consume a large number of electrical energies to transport fresh air throughout the tunnel complex. Mine ventilation accounts for 25%–60% of the total mining operating costs [1, 2], and construction ventilation accounts for up to 75% of underground construction costs. Therefore, there is a need for stable and economical construction ventilation-on-demand (VOD) systems to ensure sustainable tunnel construction. However, the branch resistance and wind volume of the ventilation-system change with construction space expansions, ventilation valve adjustments, thermal evolutions of the air environment, equipment activities, and fire accidents. Thus, an intelligent VOD systems with is urgently needed to realize the closed-loop control based on comprehensive sensing, real-time analysis, real-time control, and continuous optimization.

With increasing energy cost and fragility and complexity of the ventilation systems, VOD has attracted significant attention [3, 4]. In addition, there is a need for cost-effective ventilation systems that simultaneously minimize energy use and ensure a safe environment during tunnel group construction. During the past decade, intelligent ventilation has witnessed rapid development. The thinking, experience, and behavior of manual ventilation operations have been transformed into computational analysis and learning models, and advanced information and communication technology, equipment manufacturing, and project management methods have been applied to closed-loop ventilation control systems that could perform comprehensive sensing, real analysis, real-time control, and continuous ventilation optimization. Intelligent ventilation mitigates the problems associated with various construction procedures conducted simultaneously at different points, time-varying tunnel properties, nonlinear wind flow changes, and strong coupling of influencing factors for ventilation systems in tunnel construction.

An environmental monitoring system is referred to uses limited and well-deployed sensors to automatically record online pollutant concentrations and ventilation parameters. The accuracy and frequency of the collected data are influenced by the measurement conditions. Possible data errors of the collected data may come from airflow turbulence characteristics, nonstandard operation of the measurement equipment, construction space expansions, ventilation valve adjustments, thermal evolutions of the air environment, personnel and equipment activities, as well as fire accidents, all of which can change the wind resistance [5, 6], and, in turn, then change the wind volume [7–9]. For deviations in the monitoring data caused by the monitoring location, the relationship between the monitoring velocity and the mean velocity has been established, and the distribution of the mean velocity on the roadway section could be explored using laser Doppler velocimetry and Fluent software (ANSYS, Inc. Pittsburgh, PA, USA) [10]. Factors for correcting velocities measured at a fixed location during real-time airflow monitoring to the cross-section mean velocity have been proposed [11], and the effects of inlet velocity, ceiling roughness, and tunnel height on the cross-section velocity could be determined using Fluent software [12]. To correct monitoring data errors, experiments have shown that the turbulence data represented by mine ventilation data were consistent with Gaussian noise [13]. A computation–feedback–correction mechanism for the accurate correction of wind resistance data obtained during ventilation has been proposed to calculate roadway airflow [14]. An adaptive Kalman filter based on an expectation-maximization algorithm was proposed to shield fluid pulsations while preserving system-induced fluctuations [15]. For sensors, a multiparameter environmental monitoring system has been developed for coal mines, which allowed multiple environmental parameters to be independently monitored and processed [16]. Considering the effect of air pressure variations on air density in coal mines, a system that simultaneously monitors air pressure and wind speed has been developed [17]. After that, an ultrasonic wind-speed sensor with temperature compensation has also been developed to improve measurement accuracy [18]. Numerical simulations, such as computational fluid dynamics (CFD), have been employed to predict tunnel environments [19]. However, it is difficult to employ such simulation tools for online control because of the limitations of calculation time and identification of pollution source location and intensity. To improve the prediction speed, techniques, such as those reducing the computational grid or neglecting the Navier–Stokes equations, have also been explored. However, the prediction accuracy of these methods is compromised, especially for complex and dynamic environments. For example, fast fluid dynamics (FFD) has been proposed to quickly predict environments. Although it is approximately 50 times faster than pure CFD, it cannot perform high-precision calculations of pollutant distributions [20]. To balance prediction precision and speed, an artificial neural network (ANN) combined with limited CFD cases has been proposed for real-time prediction of environmental parameters. This advanced method showed good performance with a prediction time of less than 10 ms and an acceptable deviation lower than 10% [21]. In summary, remarkable progress has been recorded in improving conventional monitoring and prediction techniques by combining multidisciplinary knowledge of the air environment, ventilation fluids, communication, and machine learning (ML). However, these monitoring methods and systems have not been effectively linked to ventilation control processes.

The most used method for ventilation system calculation and analysis is the empirical stepped adjustment method, which, however, fails to meet the requirements of the precision and dynamic air volume load [22]. More recently, ML has emerged as a potentially disruptive technology for the control and optimization of tunnel ventilation systems. It is widely used in local resistance calculations and airflow state prediction. Several studies have modeled pressure balance using parameters based on the Darcy–Weisbach formula and ML to identify unknown parameters [23]. Furthermore, a pressure balance model was developed by mathematically modeling the pressure of duct fittings and supervised learning using support vector machines to identify unknown parameters [24]. Because the working flow characteristics of a damper are not constant, a back-propagation neural-network-based damper-opening prediction model has been proposed to predict damper opening using static pressure and demand airflow as inputs [25]. A genetic algorithm coupled with a steady-state mine ventilation network program has been extended to solve free-flow splitting optimization problems without allowing regulators [26, 27]. Convolution neural networks (CNNs) have been employed to forecast nonlinear quantities, such as airflow and air quality in underground working areas [28]. A prototype web system based on a unified geographical information system (GIS) ventilation network model was developed to provide fast simulation results and location-based information for unsteady ventilation conditions [29]. In conclusion, ML techniques have been extensively employed in local resistance calculation and parameter prediction of ventilation systems; however, the accuracy of airflow control and adaptability to the topology of different ventilation networks have not been validated.

The tunnel group ventilation control system is a large-scale complex-related system comprising intelligent control theories, effective adjustment models, and sophisticated equipment [30]. For multiobjective optimization, ML is used to resolve the trade-off between air quality and cost [31], and different formulations have been proposed [32, 33]. Automatic control software for branch VOD has been developed to calculate the ventilator frequency, branch selection, and ventilation resistance regulation using the branch parameter calculation model and ventilator network stability judgment method [34]. However, considering the stability of ventilation control systems, to further improve the accuracy of regulation control, there is a need to understand how the actions of ML agents are described and how they impact various optimization targets in multizone tunnels. In addition, how to effectively implement dynamic ventilation control systems in a tunnel construction environment is still unclear. Furthermore, there is a need to combine multidisciplinary knowledge and techniques to apply artificial intelligence techniques to engineering practice.

With the continuous improvement of environmental standards in tunnel construction, intelligent ventilation systems have become crucial for ensuring construction safety. Owing to advancements in information technology, closed-loop control technology has been implemented in various construction sites [35]. For example, for intelligent pipe-cooling control in massive concrete [36], a short-term temperature forecast model for mass concrete cooling control was developed using ANN [37]. An intelligent grouting control system named the intelligent grout control hub (iGH) has been established to achieve one-button closed-loop intelligent control of cement grouting by developing robust grouting control models, novel grouting facilities, and intelligent online monitoring approaches [38]. For intelligent security management, a full participation flat closed-loop (FPFCL) safety management method has been developedfor offshore wind power construction sites, in which the closed-loop mechanism ensures that a safety hazard was timely rectified [39]. Herein, we propose a closed-loop control research framework to ensure the supply of air demanded during tunnel group construction. Firstly, the methodology for calculating real-time ventilation load was proposed by utilizing comprehensive air environment monitoring indices, a ventilation network-based control model for analyzing the parameters of the ventilation system's network structure was established, and a reinforcement learning (RL) optimization algorithm was employed to identify the optimal state parameters for ventilation equipment and devices that met ventilation requirements, thus forming a closed-loop feedback control. Furthermore, we developed an end-edge-cloud-based closed-loop control system including data collection, data processing, real-time computation, three-dimensional (3D) visualization, and control modules. Thirdly, a software platform was developed supporting multiend ventilation data management and information sharing. The steps of this method and its verification in practical projects are described in the subsequent sections.

2 Intelligent ventilation control method

The air volume load in every area of construction ventilation undergoes dynamic changes over time, and operating the ventilation system at full load conditions can cause excessive airflow when the air load is reduced, resulting in energy wastage. Moreover, the parameters of a ventilation system are influenced by factors, such as air duct layout, fan operation parameters, and valve openings. The interdependence of these factors poses challenges for the efficient adjustment of airflow volume, resulting in prolonged adjustment times. Therefore, this section has adapted the conventional constant volume ventilation system design to accommodate the most unfavorable operating conditions, with integrated monitoring data serving as the basis for establishing control requirements of on-demand ventilation, minimum energy consumption, and real-time feedback. Therefore, the recommended fundamental criteria for an intelligent process risk control system are as follows:

(1) Demand-driven ventilation design. In order to achieve a balance between air supply and demand, a comprehensive monitoring system for the air index is established within the working area. It is required that the airflow volume for any ventilation point during each period does not deviate from the calculated load based on the monitored air index value, thus preventing inadequate or excessive ventilation.

(2) Minimum energy consumption for ventilation. A ventilation system model is established, which includes establishing the corresponding relationship between the ventilation load of monitoring points and the air volume of the ventilation network, minimizing the resistance of each branch as the goal, following the three fundamental laws, and calculating the parameters of ventilation network.

(3) Real-time feedback. The deep deterministic policy gradient (DDPG) RL algorithm is employed to dynamically control the operating parameters of ventilation equipment and devices in response to deviations between air monitoring indices and target values, thereby preventing the monitored air index data from exceeding the threshold range of the ventilation control system.

2.1 Air supply based on real-time ventilation demand

Tunnel construction involves various activities, such as excavation and drilling, blasting, smoke ventilation, safety treatment, slag transportation, shotcrete support, and reinforced concrete construction. The concentration of pollutants generated during these activities varies with their intensity and duration. As an example, Fig. 1 shows the fluctuation of dust concentration at different construction stages of a tunnel for a hydropower station. The dust concentration peaks during blasting activities and significantly drops during reinforced concrete construction, showing a concentration difference of up to 20 times. Therefore, the wind load varies dynamically over time. Operating a ventilation system at its maximum capacity would result in excessive airflow when only partial loads are present, resulting in energy inefficiency. Therefore, it is crucial for the ventilation system to dynamical calculate the ventilation load based on the monitoring concentration value of the working area.

With CO serving as the designated pollutant indicator, the factors contributing to changes in CO concentration include source emissions, ventilation, and diffusion towards non-controlled areas due to concentration gradients. According to the principle of mass conservation, the reduction in CO emissions from the control region is

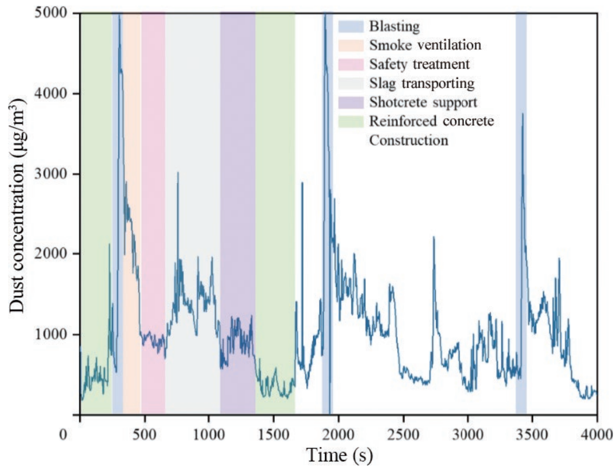


Fig. 1 Variation in pollutant concentration during tunnel construction.

equivalent to the increase in CO emissions from the non-control region, as depicted by Eq. (1)

$$V_{\mathrm{con}}\frac{\mathrm{d}C}{\mathrm{d}t} = C_{\mathrm{sou}} + Q_{\mathrm{air}}C_{\mathrm{in}} - Q_{\mathrm{air}}k(C - C_{\mathrm{non}}) \tag{1}$$

where V_{con} is the volume of the control region, Q_{air} is the ventilation quantity per unite time, k is the comprehensive diffusion coefficient of CO from the control region to the non-control region, t is the ventilation time, C is the CO concentration in the control region at time t, C_{non} is the CO concentration in the non-control region, C_{sou} is the unit time production of CO source, and C_{in} is the CO concentration of the fresh air within the air duct.

Taking the pressure-out ventilation mode after blasting as an example, it can be simplified as a one-time pollution event without internal CO source ($C_{\mathrm{sou}} = 0$) and a full fresh air supply mode without return air ($C_{\mathrm{in}} = 0$), where only the ventilation factor influences the change in CO concentration of the control region. According to the press-inventilation model of CO concentration [40], the dilution equation can be described as Eq. (2)

$$C = \lim_{n\to\infty} C_0\left(\frac{V}{V + Q_{\mathrm{air}}\Delta t}\right)^{\alpha} = \lim_{\Delta t\to 0} C_0\left(\frac{V}{V + Q_{\mathrm{air}}\Delta t}\right)^{\frac{t}{\Delta t}} = C_0\mathrm{e}^{-\frac{Q_{\mathrm{air}}t}{V}} \tag{2}$$

where the dilution volume ($V = L_0S$). L_0 is the distance between the airvent and the working face, and S is the tunnel cross sectional area. Pollutants spread rapidly once on blasting and the gas in volume V is perfectly mixed. t is divided into α time intervals ($\Delta t = t/\alpha$). $Q_{\mathrm{air}}\Delta t$ represents the air supply volume at each Δt. C_0 is the initial CO concentration.

Considering the overall distribution of CO concentration, a control volume (V_t) is established with a model length (L_t) along the tunnel axis from the working face, which includes V_{I}, V_{II}, and V_{III}, as shown in Fig. 2. V_{I}

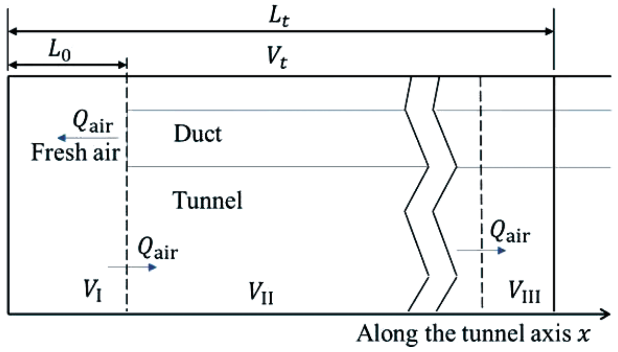


Fig. 2 Press-inventilationcalculationmodel.

represents the fully mixed zone where position $x \leqslant L_0$ and exhibits a uniform concentration distribution [39]. V_{II} denotes the one-dimensional motion region of airflow along the axis where $L_0 < x \leqslant L_0 + \frac{Q_{air}t}{S}$, and the concentration peak gradually advances. V_{III} refers to the region where the concentration peak has not yet reached, i.e., $L_0 + \frac{Q_{air}t}{S} < x < L_t$, while maintaining normal concentration levels (C_{out}).

At time t the CO concentration distribution in model volume V_t can be described as Eq. (3)

$$C_{x,t} = \begin{cases} C_{out} + (C_{x,t_i} - C_{out})e^{-\frac{kQ_{air}t}{V_I}}, \ x \leqslant L_0 \\ C_{out} + (C_{x,t_i} - C_{out})e^{-\frac{kQ_{air}\left[t - \frac{(x-L_0)S}{V_I}\right]}{V_I}}, \ L_0 < x \leqslant L_0 + \frac{Q_{air}t}{S} \\ 0, \ L_0 + \frac{Q_{air}t}{S} < x < L_t \end{cases} \tag{3}$$

where $C_{x,t}$ represents the concentration of CO at time t from position x on the working face. The monitoring device is installed in the control area V_{II}, where t_i signifies the initial ventilation time, and t_j indicates the expected end time of ventilation. Moreover, C_{x,t_i} denotes the CO concentration measured at time t_i. Q_{air} based on monitoring index within the region of $L_0 < x < L_0 + \frac{Q_{air}t}{S}$ is a constant value and can be calculated by Eq. (4)

$$Q_{air} = \frac{S}{k(t_j - t_i)}\left[x + L_0\left(\ln\frac{C_{x,t_i} - C_{out}}{C'_{x,t_j} - C_{out}} - 1\right)\right] \tag{4}$$

where C'_{x,t_j} represents the predicted concentration value at time t_j.

2.2 Ventilation control model

A detailed physical model of the ventilation control system as a ventilation network comprising fans, dampers, and ducts is developed, and a network graph $A(m, n)$ is defined as a structure consisting of m nodes and n branches, as shown in Fig. 3. We define a node as a connection point between two or more devices and accessories numbered n_1–n_9. The connections between nodes are represented by edges, with solid lines numbered e_1–e_8 connected by air ducts, while dashed lines e_9 and e_{10} represent connections through the atmosphere. The air volume of nodes i_8 and i_9 is determined by on-site monitoring of the pollution concentration, which satisfies Eq. (5). The air volume of nodes i_1–i_5 represents calculated airflow points within the ventilation network. Air volumes at nodes i_1–i_5 are determined by applying the principles of resistance, node air volume balance, and loop resistance balance to the branches connecting two nodes, and the air volume of node i_1 must satisfy both the overall air volume requirement of the ventilation network and adhere to the characteristic curve of the fan operation. Therefore, the relationship between pollutant concentration monitoring values in the control region and fan operating parameters can be determined by solving the parameters of nodes n_1–n_9 of the ventilation control model, satisfying the constraints of the nodal air volume balance, loop pressure balance, the upper and lower airflow limits of adjustable branching and fan operating frequency range, as shown in Eqs. (5)–(8)

$$\sum\nolimits_{j=1}^{n} a_{ij}q_j = 0, \ i = 1, 2, \ldots, m-1 \tag{5}$$

$$\sum\nolimits_{j=1}^{n} c_{ij}R_jq_j^2 - \sum\nolimits_{j=1}^{n} c_{ij}(P_j + F_j(q_j)) = 0, \ i = 1, 2, \ldots, n-m+1 \tag{6}$$

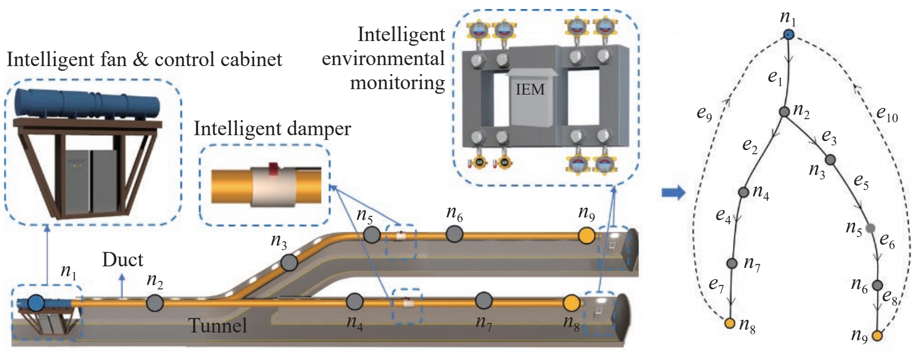


Fig. 3 Intelligent ventilation network and control model.

$$q_{j\text{-min}} \leqslant q_j \leqslant q_{j\text{-max}} \tag{7}$$

$$F_{j\text{-min}} \leqslant F_j(q_j) \leqslant F_{j\text{-max}} \tag{8}$$

where a_{ij} is an element of the basic correlation matrix, sign factor of the branch j associated with the node i, which takes the values of 1 and -1 when the air flow into and out of the node, and takes 0 when the node i is not an endpoint of a branch j. c_{ij} is an element of the basic loop matrix, sign factor of the branch j associated with the loopnode i, which take the values of 1 and -1 in the same and opposite direction, and takes 0 when the branch j is not part of loop i, q_j and R_j are the airflow and frictional wind resistance of the branch j, P_j and $F_j(q_j)$ are the natural and mechanical wind pressure in branch j, $q_{j\text{-min}}$, $q_{j\text{-max}}$, $F_{j\text{-min}}$, and $F_{j\text{-max}}$ are the lower and upper limits of the adjustable air volume and mechanical air pressure of branch j.

2.3 DDPG control algorithm

An airflow control method based on DDPG was employed to minimize ventilation energy consumption while ensuring the air volume required on demand. DDPG is a model-free and off-policy RL algorithm in which an actor–critic RL agent calculates an optimal policy by maximizing the long-term reward. In the DDPG algorithm, a copy of critic network $Q = (s, a \,|\theta^Q)$ with weight θ^Q of taking action a in state s, actor network $u = (s, a\,|\theta^u)$ with weight θ^u, where the s means and target networks Q' and u' with weights $\theta^Q \to \theta^{Q'}$ and $\theta^u \to \theta^{u'}$, respectively, are created. The agent takes an action (a_t) based on the output of the actor network to bring the environment (s_t) to the next state (s_{t+1}), as expressed in Eq. (9)

$$a_t = \pi(s_t|\theta^u) + N_t \tag{9}$$

where $\pi(s_t|\theta^u)$ is the output of the policy function representing the deterministic action calculated in state s_t, based on the current policy parameter θ^u. N_t is a random noise process to add the action exploration and the policy selected action is perturbed at each training step. The DDPG algorithm updates actors and critics at each sample time during training, whereas the agent uses an experience buffer to store past experiences. The actor and critic are then updated using a minibatch of the experiences obtained from a buffer that reduces the correlation between the samples.

To stabilize the learning process of the intelligent system, the target network weights are updated "gently" as follows

$$\theta^{Q'} = \tau\theta^Q + (1-\tau)\theta^{Q'}, \tag{10}$$

$$\theta^{u'} = \tau\theta^u + (1-\tau)\theta^{u'}, \tag{11}$$

where τ is the updated step size. $\tau \ll 1$, implying that the target value is constrained to change at a slow rate.

We define the environment of the policy as $s_t = (\theta_{1,t}, \theta_{1,t}, \ldots, \theta_{n,t}, e_{1,t}, e_{2,t}, \ldots e_{n,t})$, where $\theta_{n,t}$is the valve opening of the branch n, and $e_{n,t}$ is the difference between the actual and target airflow in the branch n. We define the action of the policy as $a_t = \Delta\theta_{n,t}$, where $\Delta\theta_{n,t}$ is the valve opening gradient of branch n at time of t and $\theta_{\min} \leqslant \theta_{n,t} + \Delta\theta_{n,t} \leqslant \theta_{\max}$, where $\theta_{n,\min} = 0$ indicates that the valve is fully opened, and $\theta_{\max} = 90^\circ$ indicates that the valve is fully closed. After that, we define the reward function (r_t) as both a penalty (c^p) and a reward (c^r) for noncompliance and conformity with the airflow standard, with the relative importance expressed in terms of β_1 and β_2 and the threshold set within 5%, as expressed in Eqs. (12)–(14)

$$r_t(s_t, a_t) = -\beta_1 c^p \max(d_{\text{ratio},t}) + \beta_2 c^r \tag{12}$$

$$c^p = \begin{cases} 0, & \max(q_{\text{ratio},t}) \leqslant E_1 \\ c_1, & E_1 < \max(q_{\text{ratio},t}) < E_2 \\ c_2, & \max(q_{\text{ratio},t}) \geqslant E_2 \end{cases} \tag{13}$$

$$c^r = \begin{cases} 1, & \max(q_{\text{ratio},t}) \leqslant E_1 \\ 0, & \text{elsewise} \end{cases} \tag{14}$$

where $d_{\text{ratio},t}$ is the maximum relative error between the real-time and target air volumes, E_1 and E_2 are the thresholds for satisfying and unsatisfying airflow requirements, c_1 and c_2 are the penalty function c^p values in distinct scenarios, with c_1 being greater than c_2.

Figure 4 shows a network structure based on a ventilation computational model, where the monitored air volume ($q_{\text{M},t}$) and pollutant concentration ($c_{\text{M},t}$), along with the target pollutant concentration ($c_{\text{tgt},t}$) at time of t, serve as inputs, the deviation of fan frequency ($\Delta f_{F,t}$) and $\Delta\theta_{n,t}$ serve as outputs. The actor and critic have three fully connected layers (FCs) with layer sizes of 128, 64, and 64. The activation function of both the critic and actor FCs is a rectified linear unit (ReLU) other than the output layers, where the action values are first passed through an FC with 64 and then superimposed behind the third FC of the critic network. The superimposed result is output after the ReLU activation function. The scaling layer is used in the output layers to standardize the output values. To train the DDPG, a minibatch size of 64 and a smoothing factor of 0.01 were used. To train RL, the Adam optimizer with an actor-network learning rate of 0.0001 and a critic network learning rate of 0.0010 were used, and L_2 regularization was employed to avoid overfitting of the neural network, with the parameter set to 0.0001. Figure 4 shows the implementation of an optimization-based filter to enforce the provided constraints. The ventilation demand of each branch is different and dynamically changes,

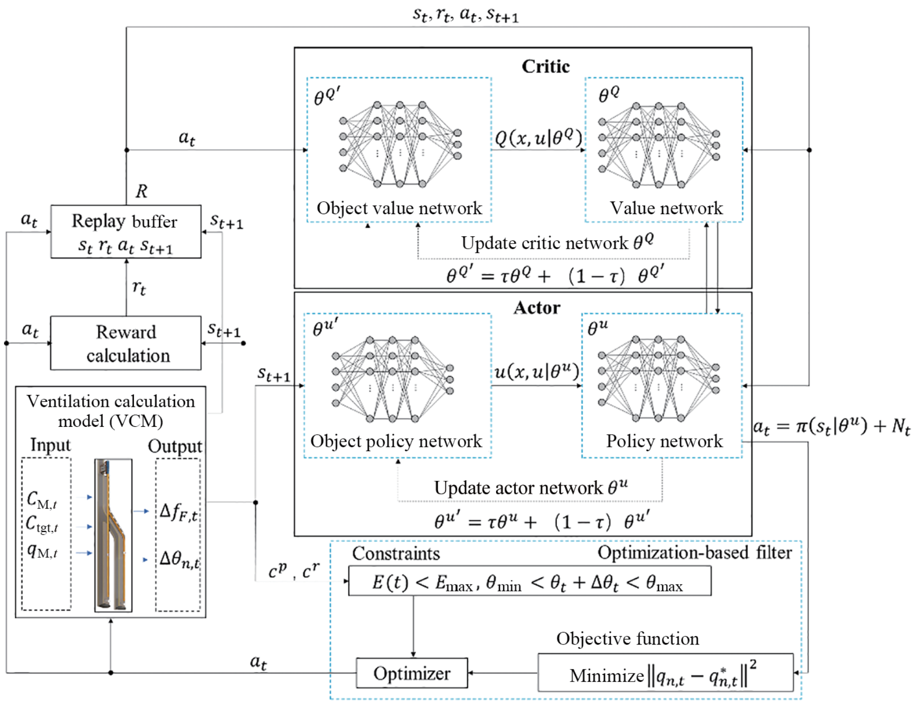


Fig. 4 Schematic of DDPG for minimizing ventilation energy consumption while meeting the required air volume.

and multiple combinations of damper openings can control the target airflow, ensuring that the discrepancy between the actual air volume $q_{n,t}$ and the targeted air volume $q^*_{n,t}$ at the time t of branch n is minimized. Therefore, a target training mechanism based on energy consumption control is employed by setting the state observed during the RL training process and the judgment condition for completing the training. The control strategy can start from any value of airflow and damper opening, $\theta_{min} < \theta_t + \Delta\theta_t < \theta_{max}$, and the total energy consumption $(E(t))$ is less than the threshold of ventilation energy consumption (E_{max}), i.e., $E(t) < E_{max}$.

2.4 Trained control strategy

The training environment is established based on the ventilation model of the underground tunnel construction ventilation system, with the objective of achieving 5 consecutive training cycles with a step length less than 20 steps for transforming the target air volume. Figure 5 shows the training process, convergence effect, and rapid adjustment of the dynamic target.

As indicated by the black solid line, the number of time steps required to complete each training cycle decreased as the number of training cycles increased. The blue solid line represents the cumulative reward obtained per training cycle. It converged toward zero as training progressed. Notably, the point at which reward convergence occurred was influenced by how rewards were set during the training.

In order to validate the regulatory capability of the trained control strategy developed through the aforementioned training, experiments were conducted in the ventilation system under varying pollutant concentrations.

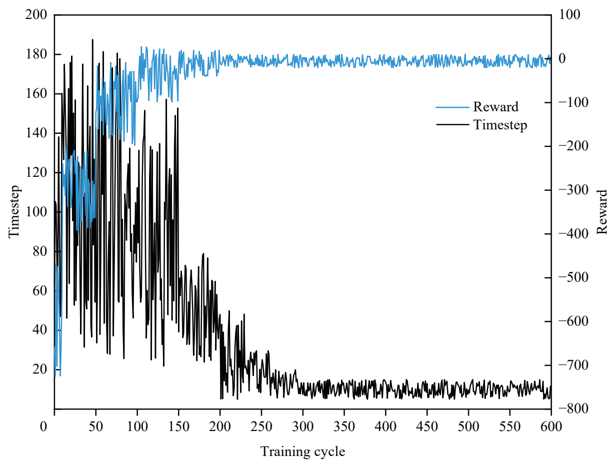


Fig. 5 Process of dynamic target training.

A total of 200 sets of target air volumes were randomly generated to assess the precision of air volume control, as depicted in Fig. 6. The black solid line represents the average relative error of the ventilation air volume, and the red dotted line indicates the permissible error threshold (7%). It can be seen that, through extensive simulation tests, the trained control strategy effectively maintained an air volume error ranging from 0 to 7%, which satisfies the criteria for applications (Apps) in ventilation projects.

3 Intelligent ventilation system

3.1 System logic

The most important concept behind the intelligent ventilation system is the logic of the closed-loop system.

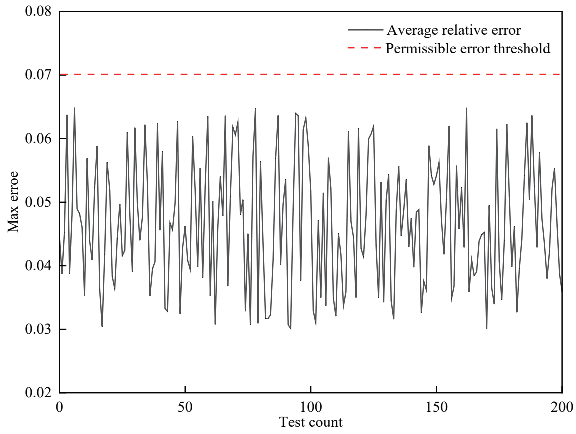


Fig. 6 Average relative error of the measured airflow.

Unlike conventional ventilation systems, this logic incorporates an interplay between ventilation efficiency, environment quality, and energy consumption. The intelligent ventilation system includes perception, calculation analysis, and adjustment of operating parameters within a closed-loop system and treats decision-making, control, and feedback as subsystems.

The closed-loop-based ventilation control cycle is iterated multiple times for continuous improvement. Closed-loop control minimizes deviations in air volume, reduces energy consumption, and ensures safety and environmentally friendly practices. After detecting the air environment parameters and forecasting the required air volume (Q_{set}), the analysis system calculates the ventilation network parameters and fan operation, thereby establishing a ventilation calculation model that expresses the relationship between environmental factors and fan operationr. The control system sets a deviation threshold (σ_{max}) for the target and actual air volumes (Q_{actual}). By ensuring $|Q_{set} - Q_{actual}| < \sigma_{max}$, on-demand ventilation functions can be provided while maintaining precise control of energy consumption. The energy consumption threshold (E_{max}) is set to regulate the usage of the ventilation system (E), i.e., $E < E_{max}$. Figure 7 shows a detailed illustration of this process, which comprises components of comprehensive perception, real-time analysis, dynamic control, and continuous optimization.

Comprehensive perception involves the use of advanced technologies, such as data acquisition and transmission technology, mobile internet, and the internet of things, to establish environmental databases (such as CO_2, CO, O_2, H_2S, CH_4, and dust particles), and state databases related to ventilation network characteristics (such as fan operation frequency (f) or damper opening degree (θ)). Data collected from diverse sources are effectively integrated and transmitted. Environmental parameter measurements enable the prediction of air volume load and serve as a foundation for calculating ventilation network parameters, whereas state parameter comparisons identify deviations from target airflow volumes. The real-time prediction of air volume load based on the mechanical-ventilation model for pollutant concentration can be determined by inputting the monitoring value of pollutant concentration, in conjunction with the target control value and ventilation time. Furthermore, real-time data visualization and early warning functions are indispensable, and management intelligence can be enhanced using large screens, personal computers (PCs), and mobile devices.

Real analysis involves computing ventilation network parameters using data from a real-time sensor. By developing a 3D program for real-time calculations, diverse information and data related to parameter variables and deviations in the ventilation system can be analyzed, evaluated, and optimized. Within the complex topology

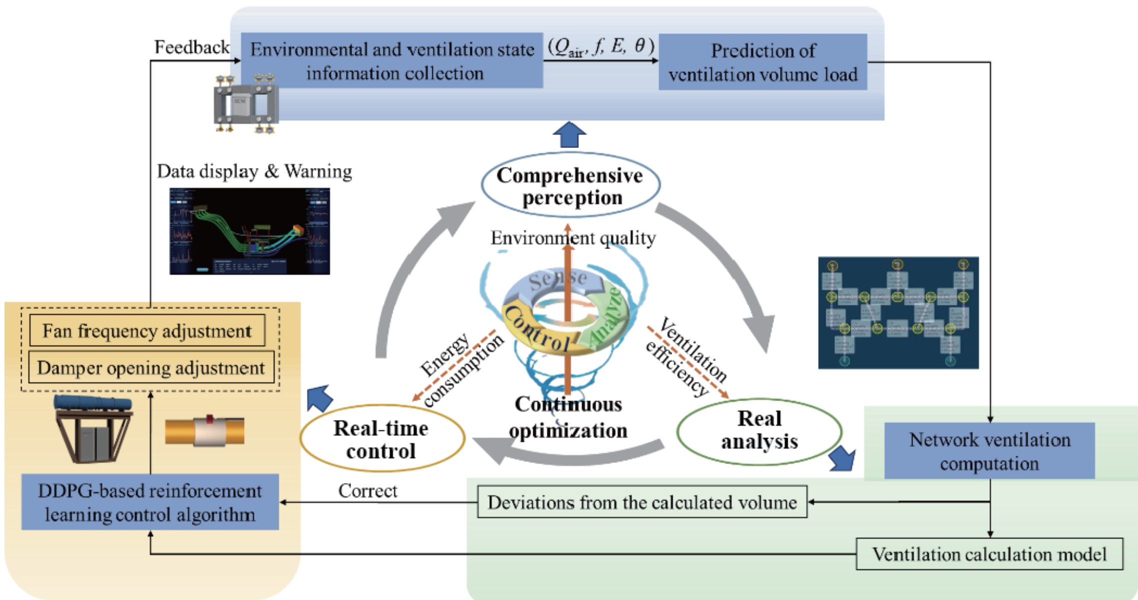


Fig. 7 Closed-loop logic for an intelligent ventilation system.

of the ventilation network, real-time solutions for branch airflow variations under various operating conditions can be determined based on input the different ventilation loads, targets and constraints. Subsequently, an updated ventilation calculation model incorporating the actual structure can be established to determine the correlation between pollutant concentration monitoring value and fan frequency and air valve opening based on ventilation network parameters calculated for the newly established ventilation network structure. This is a crucial step in updating the ventilation calculation model to serve as a foundation for DDPG real-time control. Another function utilizes the network topology to calculate the node air volume and establish a correction coefficient to enhance the accuracy of ventilation system parameter control.

Real-time control refers to the adjustment of operating parameters for the current fan and duct damper valve based on the established ventilation calculation model and deviations from the calculated air volume. Herein, a DDPG-based RL algorithm was proposed. By adjusting the exploration factor, we encourage the agent to effectively explore and address complex continuous state space problems, which aligns better with air volume control in ventilation systems. The control strategy is based on the air volume error state and establishes a reward function for minimizing deviation and the energy threshold condition for terminating optimization iterations. This approach effectively addresses the challenge of achieving rapid and precise control of air volume in multibranch ventilation systems, thereby enhancing energy efficiency and operational effectiveness.

Continuous optimization involves calculating optimal target values to enhance environmental quality, ventilation efficiency and energy consumption, thereby enabling improved capabilities for on-demand ventilation. Examples of such optimization include scheme optimization, network solution optimization, environment quality control, and energy consumption control. On-demand ventilation regulation employs a RL algorithm to improve the stability of ventilating networks when regulating fan and damper coupling methods are employed. RL employs branch air volume within these networks as states, training models to identify optimal ventilator frequencies and branch adjustment resistances that meet specified requirements. During each training phase, an initial state is obtained by resetting the environmental conditions, followed by sequential actions taken by the agents until they reach termination or a terminal state. Following each action, agents receive updated states and rewards from their environments, which inform subsequent policy updates.

3.2 System implementation

The proposed intelligent ventilation system adopts an end-edge cloud architecture, including data collection, data processing, real-time computation, 3D visualization, and control modules. Figure 8 shows the layout and implementation process of the intelligent ventilation control. The control algorithm performs edge and cloud computing. Edge computing occurs in an encapsulated integrated control cabinets with sensors for data collection and a piping and instruments diagram (PID) controller for ventilators.

The data recorded by the environmental and wind-speed sensors are acquired by the control service

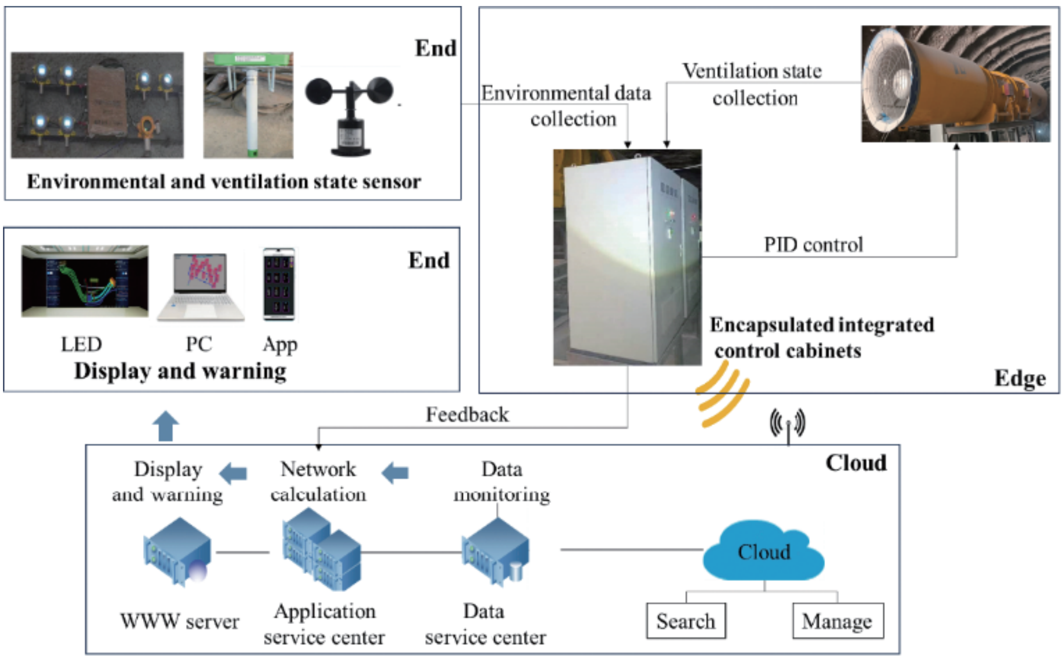


Fig. 8 Layout and implementation process of intelligent ventilation control (LED: light-emitting diode).

through the corresponding control module using the Modbus protocol. The control service then sends actuator commands and controls the valve-opening degrees to adjust the velocity. The remote-control damper is upgraded and remodeled based on the traditional damper, equipping the damper with remote automatic, local control, and full-automatic control modes, which can be switched among the functional modes. The upper computer can check the working status of each damper in real-time and send out warning signals when there is a communication interruption or failure. Cloud-based computing off-site logically comprises a message queuing telemetry transport (MQTT) service for receiving reported data and issuing commands, a web App service for PC and App functions, a database service for data storage, and a service for ML training. Edge and cloud computing are connected through the operator network.

The data collection module performs a real-time collection of air quality parameters, wind volume, and pressure data in the tunnel by deploying various sensors in the tunnel to form a sensor network. The data module processes the collected data, including data cleaning, transformation, integration, and storage. The real-time computation module computes the wind volume and pressure in the tunnel in real-time. Advanced real-time computational technology is adopted to enable a quick calculation of data in the tunnel. The 3D visualization module is responsible for 3D visualization of the computed data. It employs advanced 3D visualization technology to enable a clear display of the air status and wind pressure inside the tunnel. The control module adjusts the operational parameters and resistance of the ventilation equipment based on the computed data, achieving real-time control of the air status and wind pressure inside the tunnel.

Collaborative fusion of multisource data is also achieved by the proposed system. It integrates various data sources, including air status, wind pressure, and operational data sources, from the ventilation equipment inside the tunnel, managing and operating them through a unified data interface, thereby enabling efficient data usage and deep data mining. A system architecture is designed to support the operation of this platform. This architecture consists of data, computation, application, and user layers, each of which defines its technological connotation and main functional modules, thereby endowing the platform with excellent modularity and scalability, as shown in Fig. 9.

The data layer stores and manages various data from the tunnel, including air volume, wind pressure, and operation data from the ventilation equipment. Advanced data storage technology is employed to ensure data safety and efficiency. The computation layer computes data inside the tunnel in real-time, including data processing, computation, and analysis. Advanced real-time computation technology is employed to ensure the timeliness and accuracy of the computation. The application layer implements various application functions, including data visualization, equipment control, and alarm handling. Next generation application development technology is employed to ensure the stability and ease of use of the application functions. The user layer interacts with users, including data display, operation reception, and feedback provision. State-of-the-art user interface is employed to enhance the friendly user experience on both fixed screens and mobile devices.

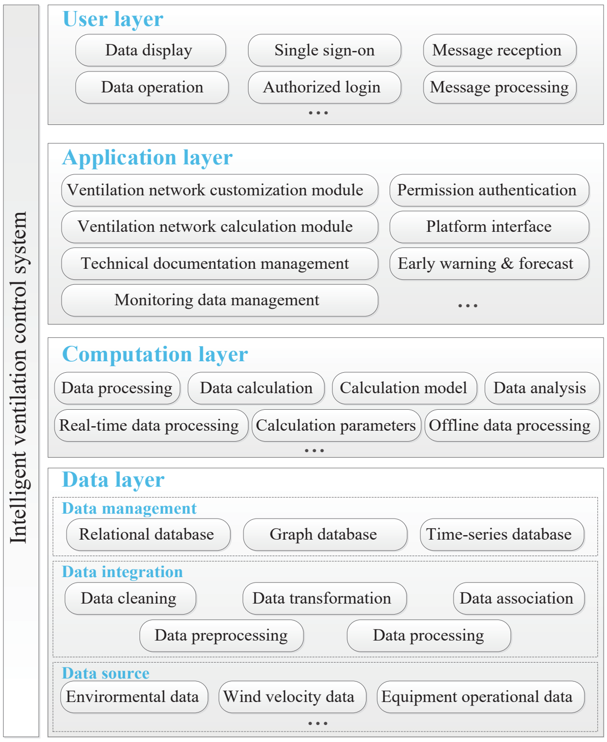


Fig. 9 Technical architecture diagram.

4 Case study

The intelligent ventilation system was implemented in the Xulong Hydropower Station in Southwest China. Ventilation effectiveness and management were optimized during the construction and maintenance periods to ensure the safe and economical operation of the ventilation system.

4.1 Xulong Hydropower Station

The analyzed regions include the main powerhouse, main transformer tunnel, tailrace surge chamber, construction area, shaft, and corridor. The dimensions of the excavation in the main powerhouse are 204.0 m (Length) × 27.7 m (Width) × 79.3 m (Height), and those of the main transformer tunnel are 175.2 m (Length) × 18.5 m (Width) × 25.0 m (Height). The transformer tunnel is connected to the main powerhouse by cable, connecting, and inlet traffic tunnels. The main transformer tunnel is also equipped with a high-voltage cable shaft and a corridor.

The tunnel group is constructed using drilling and blasting methods, and the ventilation path is complicated in the early stage, mostly in the press-out ventilation mode. It is difficult to dissipate dust and harmful gasses, which affect workers' health and safety and hamper construction. Although most of the tunnels are connected in the middle and late stages of the construction, the ventilation and dissipation of dust and harmful gasses in these stage are difficult due to the concentration of structures, high intensity of construction, construction equipment, and traffic flow, creating a good ventilation environment for concrete pouring and equipment installation. Therefore, to ensure construction progress and personnel safety, there is a need for an intelligent ventilation system to promote continuous high-intensity construction and enhance construction safety in full compatible with, structure and construction layout, and construction procedures.

4.2 Application of intelligent ventilation system

This case demonstrates ventilation during the concurrent excavation of two tunnels. The ventilation system comprises a primary fan at the entrance, which is connected to a Y-shaped air duct with one control cabinet and two branches. This pressure-based system supplies fresh air simultaneously to both working surfaces, facilitating oxygen transport and effective removal of hazardous gasses and dust particles. To accommodate the asynchronous wind load between the two branches, an intelligent control system comprising a fan control cabinet, two dampers, two sets of intelligent environmental monitors (IEMs), and three airflow meters for the air ducts was integrated into the ventilation system (Fig. 10).

The ventilation software platform was developed to incorporate 3D environmental monitoring and display, ventilation network solutions, and operation and management functionalities, as shown in Fig. 11. The 3D environmental monitoring display (Fig. 11a) shows data collected in real-time by the intelligent environmental monitoring and ventilation system in the form of curves, which are accompanied by threshold warnings, data export capabilities, and historical query options. The ventilation network calculation page (Fig. 11b) allows the construction of network nodes and branches based on the structure of the ventilation system. In the grid generation page, the attribute parameters of the nodes and branches can be set to establish the solution conditions. The designed and calculated data are stored in the network case page, where the parameters can be updated through queries or modifications. In addition, real-time calculations can be correlated with the monitoring data. The management module includes functionalities for data management and system administration. The management illustrated in Fig. 11c involves the calculation of the environmental monitoring, fan operating frequency, and energy consumption data. System administration involves log management, user management, menu management, role-based access control administration, and login control.

The development of internet technology and smart terminals has fostered the emergence of mobile-oriented ventilation control systems, empowering users with easy access and the ability to manage real-time monitoring data and fan use status through their mobile devices. Upon successful login, users can access the latest data monitored by all devices. Abnormal indicators are highlighted in red along with their corresponding threshold values. On the page of construction data input, users can select the construction site, time, and procedure. The construction procedures include excavation and drilling, blasting, smoke ventilation, safety treatment, slag transportation, shotcrete support, and reinforced concrete construction.

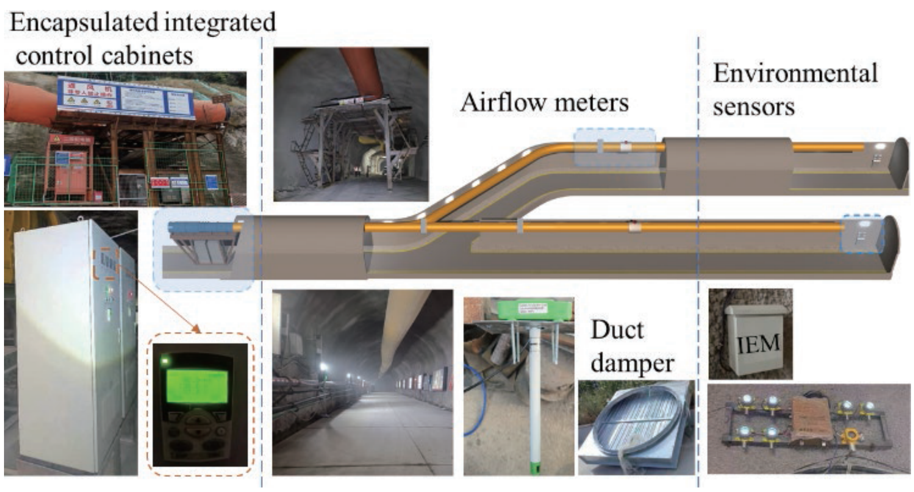
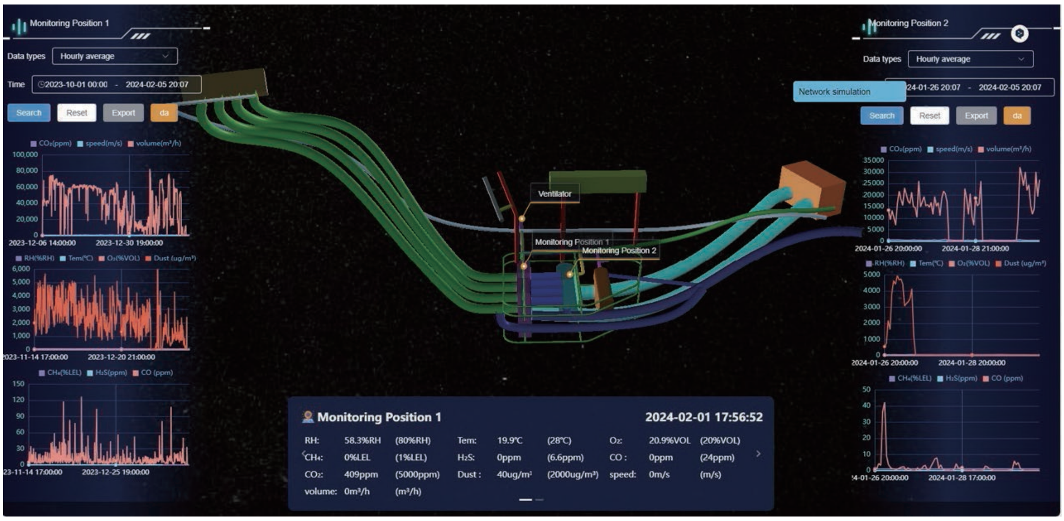
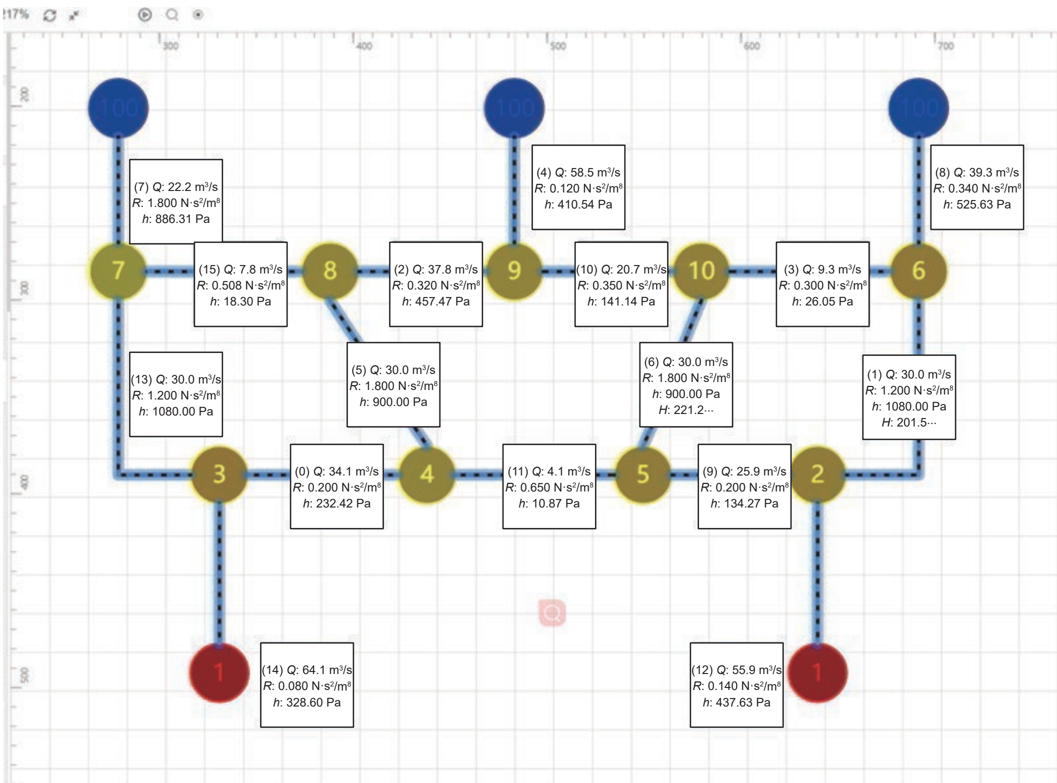


Fig. 10 Implementation of the intelligent ventilation system.



(a) 3D environmental monitoring display



(b) Ventilation network calculation

(c) Data management

Fig. 11 Ventilation software platform.

4.3 Effectiveness evaluation

Taking the blasting process as an example, the relationship among the ventilation air volume, the concentration of toxic gases and energy consumption is shown in Fig. 12. It can be seen that the CO concentration increased rapidly after blasting, reaching a maximum value of approximately 40 mg/m^3. Based on the deviation of the measured value from the target value, the fan automatically adjusted the operation frequency, which increases the ventilation air volume to a the maximum volume of about 8900 m^3/h. The fan frequency decreased after approximately 10 min of high-frequency operation, during which the CO concentration had reduced to the control value. Correspondingly, the airflow changed to 6300–7600 m^3/h, and the energy consumption significantly decreased. When the CO concentration decreased to 15 mg/m^3, the fan frequency further decreased again, and the airflow decreased to 4000–5000 m^3/h. This process continues with the next construction cycle. Thus, the intelligent ventilation system adjusts airflow on demand according to the concentration control curve and achieves the target concentration within the required time which minimizes the length of the high-frequency operation of the fan.

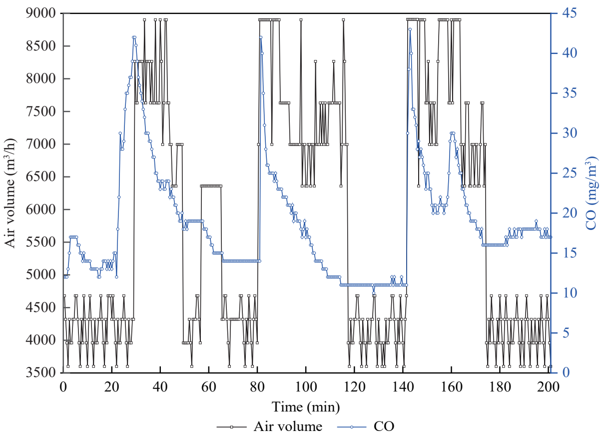
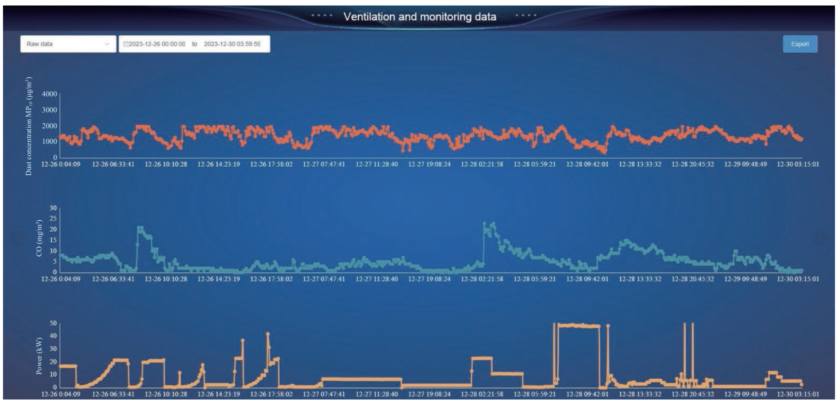
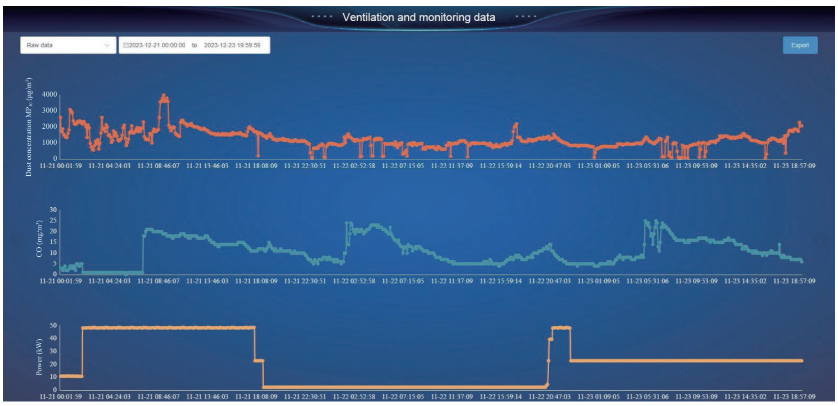


Fig. 12 Ventilation with environmental monitoring data.

The effectiveness of two ventilation control modes, namely dual-frequency control with high and low frequencies and on-demand control, was compared in terms of controlling pollutant concentrations and electricity consumption during the construction process (Fig. 13). In the dual-frequency control mode, the fan operates at a high frequency to achieve high volume ventilation once the post-blast concentration exceeds the upper threshold limit specified in the specification. Subsequently, it switches to low-frequency energy-efficient ventilation after reducing the concentration to its lowest threshold value based on empirical engineering data to ensure an adequate supply of fresh air to the work area. On the other hand, demand-driven ventilation



(a) Ventilation and operating parameters in the on-demand control model



(b) Ventilation and operating parameters in the dual-frequency control model

Fig. 13 Comparison of ventilation effectiveness and power consumption.

utilizes stepless variable frequency regulation to dynamically meet the required air supply.

Taking dust and CO as representative air quality control indicators, the monitoring data in Fig. 13 demonstrates that both ventilation control modes effectively meet the requirements for air environment. With the exception of a brief period after blasting when pollutant concentrations exceed standards, most of the time, dust (PM_{10}) concentration is maintained below 2000 μg/m^3 and CO concentration remains below 25 mg/m^3. In terms of ventilation energy consumption, the recommended fan operating range is lower 40 Hz, based on the fan characteristic curve. Beyond this frequency, there is a significant decrease in fan efficiency. The dual-frequency control mode is set at 36 and 10 Hz. In the on-demand control mode, efforts are made to maintain both motors running at the same frequency. The average energy consumption is determined based on continuous measured data, as depicted in Eqs. (15) and (16)

$$E_n = \frac{P_n + P_{n+1}}{2}(t_{n+1} - t_n) \tag{15}$$

$$\overline{E} = \frac{E_1 + E_2 + \cdots + E_n}{t_{n+1} - t_1} \tag{16}$$

where P_n and P_{n+1} are the fan power measured at the moment t_n and t_{n+1}, kW, E_n is the energy consumption of the fan during the time period from t_n to t_{n+1}, kW·h, and $\overline{E}$ represents the average energy consumption over the monitoring period from t_1 to t_{n+1}, kW·h. In the on-demand and dual-frequency control modes, the average energy consumption amounts to 8.06 and 25.63 kW·h, respectively, resulting in an energy reduction of approximately 68.54%.

5 Conclusions

In this study, an intelligent closed-loop control system for on-demand ventilation is established for the construction of complex underground tunnel groups. The following conclusions are drawn:

(1) A ventilation calculation model is established to minimizes energy consumption and an accurate representation of the ventilation system structure and parameters. Additionally, a RL optimization algorithm, DDPG, is employed to identify the optimal state parameters for ventilation equipment and devices that meet requirements.

(2) An intelligent closed-loop control ventilation system is established for the construction of complex tunnel groups, integrating comprehensive perception, real analysis, real-time control, and continuous optimization. The system employs decision-making, control, and feedback subsystems to simultaneously regulate fan frequency and damper-opening valve based on air monitoring data. It incorporates intelligent environmental monitoring devices, fan control cabinets, and smart duct dampers into conventional ventilation systems.

(3) The developed end-edge-cloud-based intelligent ventilation management platform employs the modules of a 3D environmental monitoring display, ventilation network calculation, and management. It enables early warning monitoring of environmental quality , facilitates the optimization of ventilation system design and analysis of energy consumption while supporting the management of the engineering data.

The proposed system was tested in the construction of tunnel groups construction of of Xulong Hydropower Station, validating its advanced closed-loop control based on on-demand air supply and demonstrating its applicability in engineering practices. However, further studies are needed to explore factors impacting the ventilation. Additional features can also be incorporated into the 3D visual real-time computing platform.

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Declaration of competing interest

The authors have no competing interests to declare that are relevant to the content of this article.

Author contribution statement

All authors have given approval to the final version of the manuscript.

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